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Financial Analytics Lab

GROUP 11

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Answers

[Link to EXCEL SHEET](#)

Q1. Answer to each part of question 1 is as follows

(i) We formulated 3 cases which are the present company Alpha Research, IIM X and LSE. We forecasted the cash flows of each case and discounted them according to the required discount rate to arrive at the Net Present Value of each case

NPV (Alpha Research) = INR 59772966.92

NPV (IIM X) = INR 83717694.16

NPV (LSE) = INR 82483246.88

Clearly , NPV (IIM X) > NPV (LSE) > NPV (Alpha Research)

Therefore the optimal choice is IIM X.

(ii) X: TVM of first salary as per the Geometric Progression formula

r : (1+growth rate)/(1+discount rate)

We are assuming the tax rate remains constant throughout the window

Solving in excel we get : X= INR 5644461.95

Subtracting Tax Rate we get Pre-Tax Salary as **INR 8123448.62**

(iii) Assumptions

- As soon as the job starts, repayment will be initiated
- AMMORTIZATION: X = PRINICIPAL* RATE/(1- RATE^-(REPAYMENT TIME))

Similar to part (i) we project the cash flows of two cases namely:

- (IIM X - Loan)
- (LSE - Loan)

These are notional representations for helping us analyze and incorporate the debt we have taken.

NPV(IIM X - Loan) = INR 89065051.79

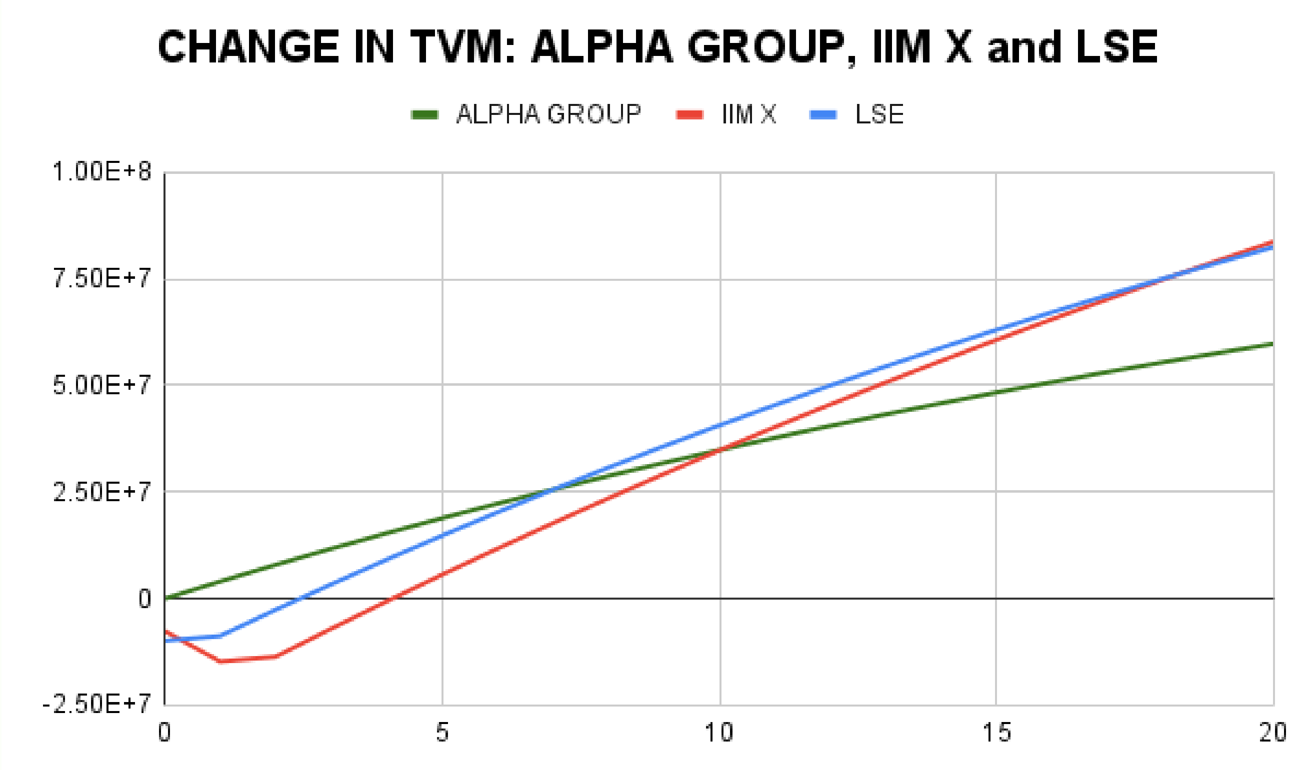
NPV(LSE - Loan) = INR 120375302.5

Clearly, NPV (IIM X - Loan) > NPV (IIM X)

NPV (LSE - Loan) > NPV (LSE)

Therefore it is clear that taking a loan is a better financial decision.

Answers



Q3. After considering different growth rates every year till year 7, we calculated the dividends based on the growth rates and discounted them to the present value. From Year 7, the constant dividend growth rate model was used to calculate the present value of the stock at Year 6, which was further discounted to get the Stock Price as ₹181.693859

	Initial Dividend	₹22.00		Present Value of dividends
D1	Year 1	-20%	₹17.60	₹15.30
D2	Year 2	-12%	₹15.49	₹11.57
D3	Year 3	-3%	₹15.02	₹9.88
D4	Year 4	15%	₹17.28	₹9.88
D5	Year 5	25%	₹21.60	₹10.74
D6	Year 6	10%	₹23.76	₹10.27
	Value of dividends of constant growth	5.50%	₹263.81	₹114.05
			Stock Price	**₹181.69**

Q4. Bond prices

- INR 1069238.18 (YTM = 5%)
- INR 761968.65 (YTM = 8%)
- INR 411589.03 (YTM = 15%)

$$\text{YTM} = \frac{\text{Cash Flow} + \dfrac{(\text{Face Value} - \text{Market Value})}{\text{Years to Maturity}}}{\dfrac{(\text{Face Value} + \text{Market Value})}{2}}$$